EXTENSION 1

Stage 6

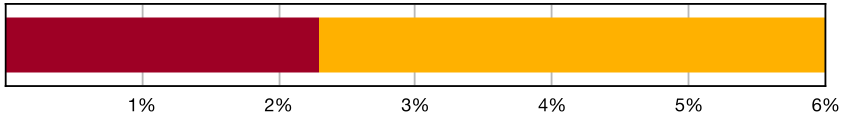
Functions (Ext1), F2 Polynomials (Y11)

Remainder and Factor Theorems (Ext1)

Teacher: SHS Maths

Exam Equivalent Time: 63 minutes *(based on allocation of 1.5 minutes per mark)*

F2 Polynomials



Remainder and Factor Theorems
Sum, Product and Multiplicity of Roots

***SmarterMaths analytics based on the average contribution to Extension 1 exams since 2020.**

HISTORICAL CONTRIBUTION

- *Polynomials* has contributed a solid 6.0%, on average, to new syllabus Ext1 exams since being introduced in 2020.
- This topic has been split into two categories for analysis purposes which are: *1-Remainder and Factor Theorems (2.3%) and 2-Sum, Products and Multiplicity of Roots (3.7%).*
- This analysis looks at *Remainder and Factor Theorems.*

HSC ANALYSIS - What to expect and common pitfalls

- *Remainder and Factor Theorems (2.3%)* has been examined in 4 out of 5 new syllabus exams (notably absent in 2024), twice in the longer answer section (2023 and 2020) and twice in the multiple choice (2022 and 2021).
- *2022 Ext1 3 MC* represents the upper difficulty level of this topic area, producing a 51% mean mark. Review carefully.
- Notwithstanding the above point, this topic area is typically examined around a band 3-4 level of difficulty and represents a great opportunity to score highly. Revise this area carefully to eliminate silly errors and get ahead of the exam clock, *in that order.*

Questions

1. Functions, EXT1 F2 2013 HSC 1 MC

The polynomial $P(x) = x^3 - 4x^2 - 6x + k$ has a factor $x - 2$.

What is the value of k?

A. 2

B. 12

C. 20

D. 36

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2. Functions, EXT1 F2 2017 HSC 1 MC

Which polynomial is a factor of $x^3 - 5x^2 + 11x - 10$?

A. $x - 2$

B. $x + 2$

C. $11x - 10$

D. $x^2 - 5x + 11$

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3. Functions, EXT1 F2 2016 HSC 2 MC

What is the remainder when $2x^3 - 10x^2 + 6x + 2$ is divided by $x - 2$?

A. -66

B. -10

C. $-x^3 + 5x^2 - 3x - 1$

D. $x^3 - 5x^2 + 3x + 1$

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4. Functions, EXT1 F2 2019 MET2-N 3 MC

If $x + a$ is a factor of $8x^3 - 14x^2 - a^2x$, where $a \in R, a \neq 0$, then the value of a is

A. 7

B. 4

C. 1

D. -2

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5. Functions, EXT1 F2 2021 HSC 3 MC

What is the remainder when $P(x) = -x^3 - 2x^2 - 3x + 8$ is divided by $x + 2$?

A. -14

B. -2

C. 2

D. 14

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6. Functions, EXT1 F2 EQ-Bank 3 MC

When $2x^3 - x^2 + px - 6$ is divided by $x + 2$ the remainder is -4. What is the value of p ?

A. -11

B. -7

C. -5

D. -2

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7. Functions, EXT1 F2 SM-Bank 2 MC

Let $p(x) = x^3 - 2ax^2 + x - 1$. When $p(x)$ is divided by $x + 2$, the remainder is 5.

The value of a is

A. 2

B. $\frac{1}{2}$

C. $-\frac{3}{2}$

D. -2

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8. Functions, EXT1 F2 2015 HSC 1 MC

What is the remainder when $x^3 - 6x$ is divided by $x + 3$?

A. -9

B. 9

C. $x^2 - 2x$

D. $x^2 - 3x + 3$

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9. Functions, EXT1 F2 2012 HSC 8 MC

When the polynomial $P(x)$ is divided by $(x+1)(x-3)$, the remainder is $2x + 7$.

What is the remainder when $P(x)$ is divided by $x - 3$?

A. 1

B. 7

C. 9

D. 13

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10. Functions, EXT1 F2 2014 HSC 9 MC

The remainder when the polynomial $P(x) = x^4 - 8x^3 - 7x^2 + 3$ is divided by $x^2 + x$ is $ax + 3$.

What is the value of a?

A. -14

B. -11

C. -2

D. 5

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11. Functions, EXT1 F2 2022 HSC 3 MC

Let $P(x)$ be a polynomial of degree 5. When $P(x)$ is divided by the polynomial $Q(x)$, the remainder is $2x + 5$.

Which of the following is true about the degree of Q?

A. The degree must be 1.

B. The degree could be 1.

C. The degree must be 2.

D. The degree could be 2.

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12. Functions, EXT1 F2 2018 HSC 11a

Consider the polynomial $P(x) = x^3 - 2x^2 - 5x + 6$.

i. Show that $x = 1$ is a zero of $P(x)$. ***(1 mark)***

ii. Find the other zeros. ***(2 marks)***

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13. Functions, EXT1 F2 2020 HSC 11a

Let $P(x) = x^3 + 3x^2 - 13x + 6$.

i. Show that $P(2) = 0$. ***(1 mark)***

ii. Hence, factor the polynomial $P(x)$ as $A(x)B(x)$, where $B(x)$ is a quadratic polynomial. ***(2 marks)***

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14. Functions, EXT1 F2 2015 HSC 11f

Consider the polynomials $P(x) = x^3 - kx^2 + 5x + 12$ and $A(x) = x - 3$.

i. Given that $P(x)$ is divisible by $A(x)$, show that $k = 6$. ***(1 mark)***

ii. Find all the zeros of $P(x)$ when $k = 6$. ***(2 marks)***

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15. Functions, EXT1 F2 2019 HSC 11d

Find the polynomial $Q(x)$ that satisfies $x^3 + 2x^2 - 3x - 7 = (x-2)Q(x) + 3$. ***(2 marks)***

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16. Functions, EXT1 F2 2023 HSC 11c

Consider the polynomial

$$P(x) = x^3 + ax^2 + bx - 12,$$

where a and b are real numbers.

It is given that $x + 1$ is a factor of $P(x)$ and that, when $P(x)$ is divided by $x - 2$, the remainder is -18.

Find a and b. ***(3 marks)***

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17. Functions, EXT1 F2 2007 HSC 2c

The polynomial $P(x) = x^2 + ax + b$ has a zero at $x = 2$. When $P(x)$ is divided by $x + 1$, the remainder is 18.

Find the values of a and b. ***(3 marks)***

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18. Functions, EXT1 F2 2009 HSC 2a

The polynomial $p(x) = x^3 - ax + b$ has a remainder of 2 when divided by $(x-1)$ and a remainder of 5 when divided by $(x+2)$.

Find the values of a and b. ***(3 marks)***

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19. Functions, EXT1 F2 2011 HSC 2a

Let $P(x) = x^3 - ax^2 + x$ be a polynomial, where a is a real number.

When $P(x)$ is divided by $x - 3$ the remainder is 12.

Find the remainder when $P(x)$ is divided by $x + 1$. ***(3 marks)***

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20. Functions, EXT1 F2 2008 HSC 1a

The polynomial x^3 is divided by $x + 3$. Calculate the remainder. ***(2 marks)***

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21. Functions, EXT1 F2 2004 HSC 3b

Let $P(x) = (x+1)(x-3)Q(x) + a(x+1) + b$, where $Q(x)$ is a polynomial and a and b are real numbers.

When $P(x)$ is divided by $(x+1)$ the remainder is -11.

When $P(x)$ is divided by $(x-3)$ the remainder is 1.

i. What is the value of b? ***(1 mark)***

ii. What is the remainder when $P(x)$ is divided by $(x+1)(x-3)$? ***(2 marks)***

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22. Functions, EXT1 F2 2010 HSC 2c

Let $P(x) = (x+1)(x-3)Q(x) + ax + b$,

where $Q(x)$ is a polynomial and a and b are real numbers.

The polynomial $P(x)$ has a factor of $x - 3$.

When $P(x)$ is divided by $x + 1$ the remainder is 8.

i. Find the values of a and b. ***(2 marks)***

ii. Find the remainder when $P(x)$ is divided by $(x+1)(x-3)$. ***(1 mark)***

Worked Solutions

1. Functions, EXT1 F2 2013 HSC 1 MC

$P(x) = x^3 - 4x^2 - 6x + k$

Since $(x-2)$ is a factor, $P(2) = 0$

$$2^3 - 4 \cdot 2^2 - 6 \cdot 2 + k = 0$$
$$8 - 16 - 12 + k = 0$$
$$k = 20$$

$\Rightarrow C$

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2. Functions, EXT1 F2 2017 HSC 1 MC

$$f(2) = 2^3 - 5 \cdot 2^2 + 11 \cdot 2 - 10$$
$$= 8 - 20 + 22 - 10$$
$$= 0$$

$\therefore (x-2)$ is a factor

$\Rightarrow A$

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3. Functions, EXT1 F2 2016 HSC 2 MC

$$P(2) = 2 \cdot 2^3 - 10 \cdot 2^2 + 6 \cdot 2 + 2$$
$$= -10$$

$\Rightarrow B$

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4. Functions, EXT1 F2 2019 MET2-N 3 MC

$$f(-a) = 8(-a)^3 - 14(-a)^2 - a^2(-a)$$
$$0 = -8a^3 - 14a^2 + a^3$$
$$0 = -7a^3 - 14a^2$$
$$0 = -7a^2(a+2)$$
$$a = -2$$

$\Rightarrow D$

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5. Functions, EXT1 F2 2021 HSC 3 MC

$$P(-2) = -(-2)^3 - 2(-2)^2 - 3(-2) + 8$$
$$= 8 - 8 + 6 + 8$$
$$= 14$$

$\Rightarrow D$

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6. Functions, EXT1 F2 EQ-Bank 3 MC

Since $(2x^3 - x^2 + px - 6) \div (x+2) = -4 \ \Rightarrow \ P(-2) = -4$

$$2 \times (-2)^3 - (-2)^2 - 2p - 6 = -4$$
$$-16 - 4 - 2p - 6 = -4$$
$$2p = -22$$
$$p = -11$$

$\Rightarrow A$

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7. Functions, EXT1 F2 SM-Bank 2 MC

$$P(-2) = 5$$
$$5 = (-2)^3 - 2a(-2)^2 - 2 - 1$$
$$5 = -8 - 8a - 2 - 1$$
$$8a = -16$$
$$\therefore a = -2$$

$\Rightarrow D$

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8. Functions, EXT1 F2 2015 HSC 1 MC

$$\text{Remainder} = P(-3)$$
$$= (-3)^3 - 6(-3)$$
$$= -27 + 18$$
$$= -9$$

$\Rightarrow A$

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9. Functions, EXT1 F2 2012 HSC 8 MC

Let $P(x) = A(x) \cdot Q(x) + R(x)$

where $A(x) = (x+1)(x-3)$, and $R(x) = 2x+7$

When $P(x) \div (x-3)$, remainder $= P(3)$

$$\begin{aligned} P(3) &= 0 + R(3) \\ &= (2 \times 3) + 7 \\ &= 13 \end{aligned}$$

$\Rightarrow D$

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10. Functions, EXT1 F2 2014 HSC 9 MC

$P(x) = x^4 - 8x^3 - 7x^2 + 3$

Given $P(x) = \left(x^2 + x\right) \cdot Q(x) + ax + 3$

$= x(x+1)Q(x) + ax + 3$

$P(-1) = 1 + 8 - 7 + 3 = 5$

$\therefore -a + 3 = 5$

$a = -2$

$\Rightarrow C$

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11. Functions, EXT1 F2 2022 HSC 3 MC

Given $P(x)$ has degree 5

$P(x) \div Q(x)$ has remainder $2x+5$

♦ Mean mark 51%.

Consider examples to resolve possibilities:

eg. $x^5 + 2x + 5 \div x^3 = x^2 +$ remainder $2x+5$

$\therefore$ Degree must be 2 is incorrect

$Q(x)$ can have a degree of 2, 3 or 4

$\Rightarrow D$

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12. Functions, EXT1 F2 2018 HSC 11a

i. $P(1) = 1 - 2 - 5 + 6 = 0$

$\therefore x = 1$ is a zero

ii. Using part (i) $\Rightarrow (x-1)$ is a factor of $P(x)$

$P(x) = (x-1) \cdot Q(x)$

By long division:

$$\begin{aligned} P(x) &= (x-1)\left(x^2 - x - 6\right) \\ &= (x-1)(x-3)(x+2) \end{aligned}$$

$\therefore$ Other zeroes are:

$x = -2$ and $x = 3$

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13. Functions, EXT1 F2 2020 HSC 11a

i. $$\begin{aligned} P(2) &= 8 + 12 - 26 + 6 \\ &= 0 \end{aligned}$$

ii.

$$\begin{array}{r@{\,}l} & x^2 + 5x - 3 \\ x - 2 \,\big) & \overline{x^3 + 3x^2 - 13x + 6} \\ & \underline{x^3 - 2x^2} \\ & 5x^2 - 13x + 6 \\ & \underline{5x^2 - 10x} \\ & -3x + 6 \\ & \underline{-3x + 6} \\ & 0 \end{array}$$

$\therefore P(x) = (x-2)\left(x^2 + 5x - 3\right)$

14. Functions, EXT1 F2 2015 HSC 11f

i. $P(x) = x^3 - kx^2 + 5x + 12$

$A(x) = x - 3$

If $P(x)$ is divisible by $A(x) \quad \Rightarrow \quad P(3) = 0$

$$\begin{aligned} 3^3 - k\left(3^2\right) + 5 \times 3 + 12 &= 0 \\ 27 - 9k + 15 + 12 &= 0 \\ 9k &= 54 \\ \therefore k &= 6 \quad \text{... as required} \end{aligned}$$

ii. Find all roots of $P(x)$

$P(x) = (x - 3) \cdot Q(x)$

Using long division to find $Q(x)$:

$$\begin{array}{r l} & x^2 - 3x - 4 \\ \cline{2-2} x - 3 & \big) \, x^3 - 6x^2 + 5x + 12 \\ & \underline{x^3 - 3x^2} \\ & \quad -3x^2 + 5x + 12 \\ & \quad \underline{-3x^2 + 9x} \\ & \qquad\quad -4x + 12 \\ & \qquad\quad \underline{-4x + 12} \\ & \qquad\qquad\qquad 0 \end{array}$$

$$\begin{aligned} \therefore P(x) &= x^3 - 6x^2 + 5x + 12 \\ &= (x - 3)\left(x^2 - 3x - 4\right) \\ &= (x - 3)(x - 4)(x + 1) \end{aligned}$$

$\therefore$ Zeros at $\quad x = -1, 3, 4$

15. Functions, EXT1 F2 2019 HSC 11d

$$\begin{aligned} (x - 2) \cdot Q(x) + 3 &= x^3 + 2x^2 - 3x - 7 \\ (x - 2) \cdot Q(x) &= x^3 + 2x^2 - 3x - 10 \end{aligned}$$

$$\begin{array}{r l} & x^2 + 4x + 5 \\ \cline{2-2} x - 2 & \big) \, x^3 + 2x^2 - 3x - 10 \\ & \underline{x^3 - 2x^2} \\ & \qquad 4x^2 - 3x - 10 \\ & \qquad \underline{4x^2 - 8x} \\ & \qquad\qquad 5x - 10 \\ & \qquad\qquad \underline{5x - 10} \\ & \qquad\qquad\qquad 0 \end{array}$$

$\therefore Q(x) = x^2 + 4x + 5$

16. Functions, EXT1 F2 2023 HSC 11c

$(x + 1)$ is a factor of $P(x) \quad \Rightarrow \quad P(-1) = 0$

$0 = (-1)^3 + a - b - 12$

$a - b = 13 \quad \text{... (1)}$

$P(x) \div (x - 2)$ has a remainder of $-18 \ \Rightarrow \ P(2) = -18$

$-18 = 8 + 4a + 2b - 12$

$4a + 2b = -14$

$2a + b = -7 \quad \text{... (2)}$

Add $(1) + (2)$:

$3a = 6 \quad \Rightarrow \quad a = 2$

Substitute $a = 2$ into (1):

$2 - b = 13 \quad \Rightarrow \quad b = -11$

$\therefore a = 2,\ b = -11$

17. Functions, EXT1 F2 2007 HSC 2c

$P(x) = x^2 + ax + b$

Since there is a zero at $x = 2$,

$P(2) = 0$

$2^2 + 2a + b = 0$

$2a + b = -4 \quad \ldots (1)$

$P(-1) = 18,$

$(-1)^2 - a + b = 18$

$-a + b = 17 \quad \ldots (2)$

Subtract $(1) - (2)$,

$3a = -21$

$a = -7$

Substitute $a = -7$ into (1),

$2(-7) + b = -4$

$b = 10$

$\therefore a = -7$ and $b = 10$

18. Functions, EXT1 F2 2009 HSC 2a

$p(x) = x^3 - ax + b$

$P(1) = 2$

$1 - a + b = 2$

$b = a + 1 \quad \ldots (1)$

$P(-2) = 5$

$-8 + 2a + b = 5$

$2a + b = 13 \quad \ldots (2)$

Substitute $b = a + 1$ into (2)

$2a + a + 1 = 13$

$3a = 12$

$\therefore a = 4$

$\therefore b = 5$

19. Functions, EXT1 F2 2011 HSC 2a

$P(x) = x^3 – ax^2 + x$

Since $P(x) \div (x – 3)$ has remainder 12,

$P(3) = 3^3 - a \times 3^2 + 3 = 12$

$27 - 9a + 3 = 12$

$9a = 18$

$a = 2$

$\therefore\ P(x) = x^3 - 2x^2 + x$

Remainder $P(x) \div (x + 1)$ is $P(–1)$

$P(-1) = (-1)^3 - 2(-1)^2 - 1$

$= –4$

20. Functions, EXT1 F2 2008 HSC 1a

$P(-3) = (-3)^3$

$= -27$

$\therefore$ Remainder when $x^3 \div (x + 3) = -27$

MARKER'S COMMENT: "Grave concern" that many who found $P(-3) = -27$ stated the remainder was 27.

21. Functions, EXT1 F2 2004 HSC 3b

i. $P(x) = (x + 1)(x - 3)Q(x) + a(x + 1) + b$

$P(-1) = -11$

$-11 = (-1 + 1)(-1 - 3)Q(x) + a(-1 + 1) + b$

$-11 = (0)(-4)Q(x) + a(0) + b$

$\therefore b = -11$

(ii) $P(3) = 1$

$\therefore (3 + 1)(3 - 3)Q(x) + a(3 + 1) - 11 = 1$

$4a = 12$

$a = 3$

When $P(x)$ is divided by $(x + 1)(x - 3)$

$R(x) = a(x + 1) + b$

$= 3(x + 1) - 11$

$= 3x + 3 - 11$

$= 3x - 8$

22. Functions, EXT1 F2 2010 HSC 2c

i. $P(x) = (x+1)(x-3)Q(x) + ax + b$

$(x-3)$ is a factor (given)

$\therefore P(3) = 0$

$3a + b = 0 \quad \ldots (1)$

$P(x) \div (x+1) = 8 \quad \text{(given)}$

$\therefore P(-1) = 8$

$-a + b = 8 \quad \ldots (2)$

Subtract $(1) - (2)$

$4a = -8$

$a = -2$

Substitute $a = -2$ into (1)

$-6 + b = 0$

$b = 6$

$\therefore a = -2,\ b = 6$

ii. $P(x) \div (x+1)(x-3)$

$$= \frac{(x+1)(x-3)Q(x) - 2x + 6}{(x+1)(x-3)}$$

$$= Q(x) + \frac{-2x+6}{(x+1)(x-3)}$$

$\therefore$ Remainder is $-2x + 6$

COMMENT: This question requires a fundamental understanding of the remainder theorem.